SE4041 MADD • ASSIGNMENT 02 (PART B: TVOS)

# PartyPlay TV

## Your Living Room Game Show

Native Apple tvOS Multiplayer Trivia Platform


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**MODULE:** SE4041 – Mobile Application Design & Development

**DEGREE / FACULTY:** BSc (Hons) in Information Technology • SLIIT

**TECH STACK:** tvOS 17.2 SDK • Swift 5.9 • Apple TV 4K

tvOS 17.2 | Focus Engine | CoreImage QR | 100% Offline Demo




### Living Room Broadcast Stage

Transforms the Apple TV 4K into an interactive game show host. Controllable via the physical Siri Remote or companion mobile buzzers.

PROBLEM STATEMENT

# The Living Room Entertainment Disconnect

4K television screens are massive, yet shared living room play is missing.

THE SCREEN

## Passive Video

4K TVs mostly play video

- Families sit together but watch passively
- The TV acts as a receiver, not a game host
- Little content invites shared doing and play

THE HABIT

## Small-Screen Silos

Mobile trivia isolates players

- Everyone looks down into personal phone screens
- Kills the communal living-room atmosphere
- Lacks the thrill of a shared game show

THE SETUP

## Costly & Complex

Console party games have friction

- Jackbox packs cost $30+ each
- Requires multiple gamepads or cloud servers
- No free, native offline Apple TV alternative

**THE GAP: Turn the living room's centerpiece screen into an instant, social party game show.**

PROPOSED SOLUTION

# A Live Game Show on Your TV Screen

PartyPlay TV turns the Apple TV into an interactive game-show stage with zero setup friction.

**THE MAIN IDEA: Turn the living-room TV into a broadcast game show you play with a remote.**

### 1. NATIVE 10-FOOT FOCUS UI

Built for the couch with SwiftUI @FocusState, parallax card elevation (1.06x), and glowing cyan halos.

### 2. DUAL-MODE FLEXIBILITY

100% offline with smart AI bots for solo demo play, plus instant CoreImage QR codes for phone buzzers.

### 3. LIVE GAME ENGINE

Real-time countdown ring, dynamic +50% speed bonuses, instant educational feedback, and podium rankings.

### SOLUTION ARCHITECTURE

▶ **HOST SCREEN (Apple TV 4K)**
Broadcasts questions, circular timer & podium

▶ **INTERACTION MODES**
Siri Remote Focus Engine • CoreImage QR Handshake

▶ **CONTESTANT ROSTER**
Solo Remote Host • Simulated AI Bots • Phone Buzzers

▶ **SWIFT MVVM CORE**
GameEngine (Timer/State) • ScoreService (Bonuses)

VALUE PROPOSITION & KEY FEATURES

# Fast Setup, Fierce Competition

Instant couch entertainment designed for players of all ages.

10-FOOT DESIGN

## Siri Remote First

- Parallax card lift (1.06x)
- Glowing cyan focus halos
- Readable from 3+ meters
- Tactile sound feedback

CURATED TRIVIA

## 6 Categories

- 42 verified questions
- Science, Gaming, Movies
- Sri Lanka Quiz heritage
- Easy, Medium, Hard tiers

REACTIVE ENGINE

## Dynamic Scoring

- Circular countdown ring
- Up to +50% speed bonus
- Anti-spam submission lock
- Tie-breaker rank logic

ZERO FRICTION

## 100% Offline Ready

- Works with zero internet
- Simulated AI bots included
- Native CoreImage QR codes
- Zero accounts, zero ads

APPLICATION OVERVIEW & USER FLOW

# One Clean Path Through 6 Screens

A smooth linear journey powered by SwiftUI NavigationStack and AppRoute routing.

STAGE 1
**Home**
Hero branding, Play Now & recent match summary.

STAGE 2
**Categories**
6 trivia themes with custom palettes & difficulty tags.

STAGE 3
**Setup**
Select timer (10-20s), question count & tier.

STAGE 4
**Lobby**
Room code PLAY88, QR code & AI bot roster.

STAGE 5
**Quiz Arena**
Countdown ring, 2x2 answers & instant reveal.

STAGE 6
**Podium**
Gold/Silver/Bronze pedestals & rankings table.

## SIRI REMOTE CONTROL GESTURES

- **Touchpad Swipe:** Glide focus between category cards and answer choices with native spring physics.
- **Clickpad Select:** Confirm selection, submit answer choice, and advance questions.
- **Menu / Back Button:** Pop cleanly back up the navigation stack without losing game session state.

FEATURE SHOWCASE I

# 10-Foot Design & Siri Remote Focus

Television-calibrated layout, parallax card scaling, and tactile focus states.

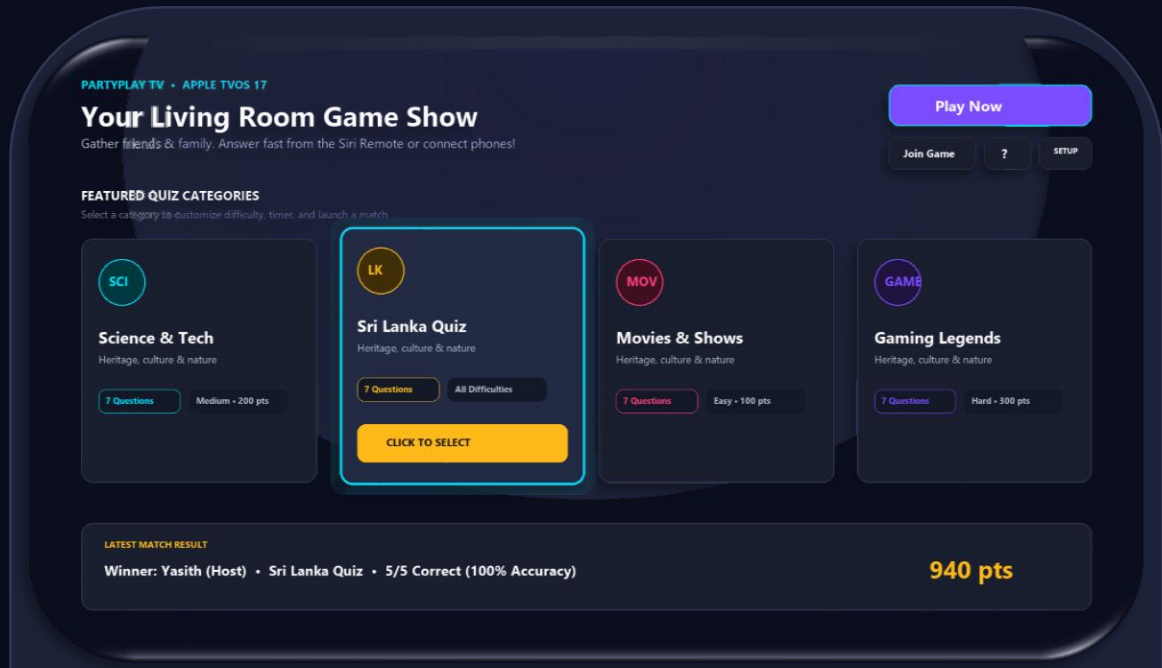
## FOCUS ENGINE ARCHITECTURE

**▶ Apple tvOS HIG Standard**
90pt TV-safe margins with distance typography readable from 3+ meters away.

**▶ @FocusState Management**
Pure SwiftUI focus orchestration without mouse pointers or touchscreens.

**▶ Parallax Card Elevation**
Focused cards lift 1.06x with a bright cyan glowing border (#00E5FF).

**▶ Sri Lanka Quiz Spotlight**
Dedicated category featuring verified national history, geography, and sports.

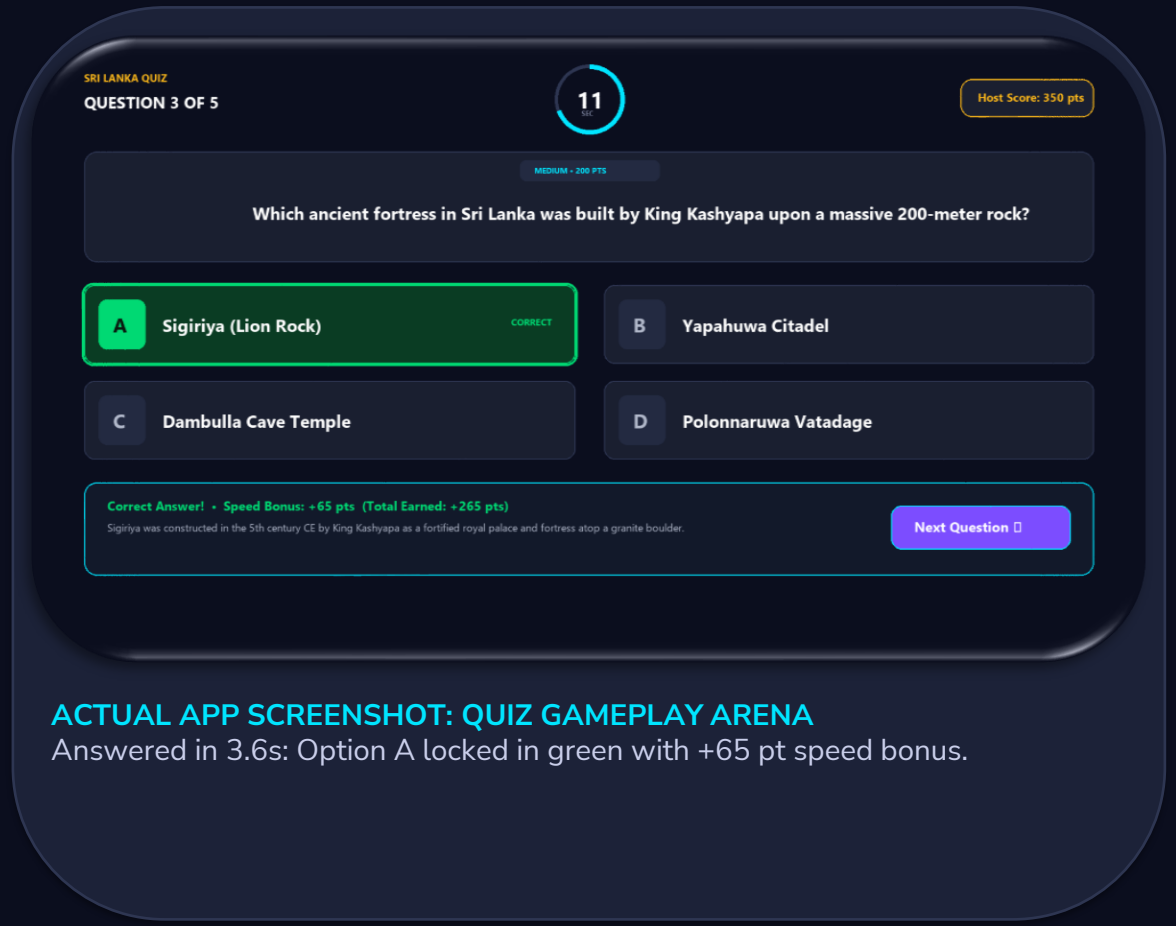



**ACTUAL APP SCREENSHOT: HOME & CATEGORY SELECTION**
Focused card lifts with parallax scale and cyan neon glow.

FEATURE SHOWCASE II

# Real-Time Timer & Speed-Bonus Scoring

Broadcast question arena with synchronized countdown, speed bonuses, and educational feedback.




**ACTUAL APP SCREENSHOT: QUIZ GAMEPLAY ARENA**
Answered in 3.6s: Option A locked in green with +65 pt speed bonus.

## SCORING & FEEDBACK MECHANICS

**▶ Synchronized Countdown Ring**
Color shifts Cyan (>50%) ➔ Amber (>20%) ➔ Red (<20%).

**▶ Speed-Bonus Algorithm**
Formula: Score = Base + (Base * 0.5 * TimeLeft / TotalTime).

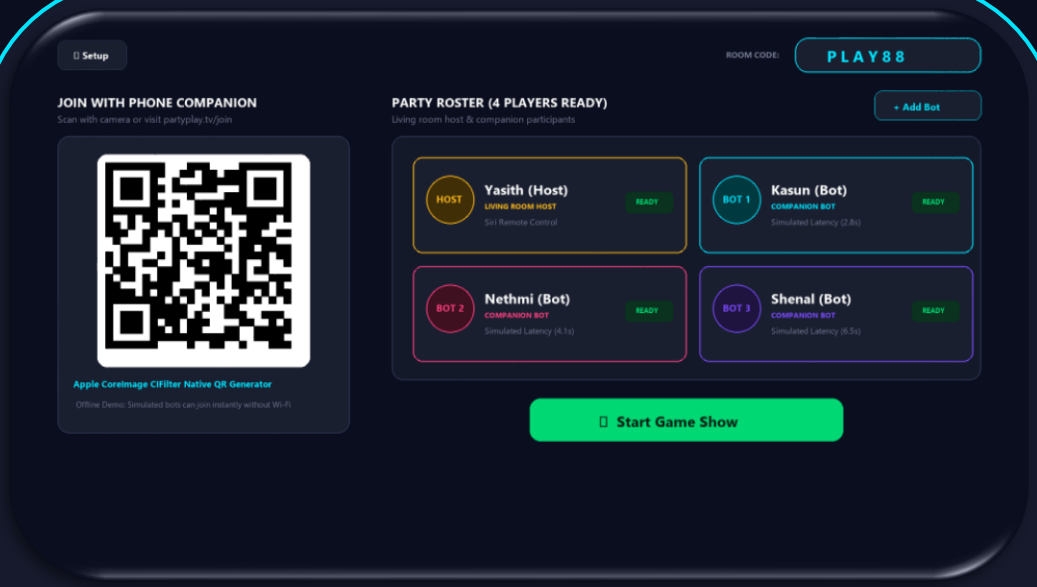
**▶ Anti-Spam Input Lock**
State machine locks on initial click to prevent double submissions.

**▶ Instant Educational Reveal**
Displays verified historical context and auto-focuses 'Next Question'.

FEATURE SHOWCASE III

# Simulated Bot Lobby & Winner Podium

Dual-mode multiplayer with native CoreImage QR pairing and deterministic podium rankings.




## COREIMAGE QR & OFFLINE BOTS

- Native CoreImage QR: Encodes companion URLs without external libraries.
- Simulated AI Bots: Bots simulate human response delays (2–15s) for instant offline play.




## CHAMPION PODIUM CELEBRATION

- 3D-Style Podium: Gold (#1, 250pt), Silver (#2, 190pt), Bronze (#3, 150pt).
- Deterministic Tie-Break: Score ➔ Correct Count ➔ Accuracy %.
- JSON Storage: Archives match history to Application Support.

TARGET AUDIENCE & MARKET

# Capturing the Shared Living Room

Targeting multi-generational families, university social groups, and smart classrooms.

PRIMARY TARGET

## Families

Wholesome shared screen time

- Grandparents & kids play together
- Safe, educational heritage trivia
- Replaces silent phone scrolling

SECONDARY TARGET

## University Dorms

Fast-paced party gaming

- Social game nights & dorm tournaments
- Competitive speed-bonus tension
- Zero cost, no console required

TERTIARY TARGET

## Classrooms

Interactive visual quizzes

- AirPlay quizzes on lecture displays
- High legibility from back of room
- Verified educational explanations

**MARKET OPPORTUNITY & DATA METRICS**

- **40M+ Active Apple TV Devices:** Premium living-room household base with 4K screens (Statista / Apple).
- **$18.2B Party Gaming Market:** Growing at 8.4% annually with high demand for co-located social play (Newzoo).
- **Blue Ocean Opportunity:** Zero direct competitors offering localized Sri Lankan trivia on Apple tvOS.

DIFFERENTIATION & ROADMAP

# Why PartyPlay TV Wins & Future Roadmap

Clear advantages over existing alternatives, backed by a phased expansion plan.

| Feature | PartyPlay TV | Jackbox | Kahoot! | TV Clones |
|---|---|---|---|---|
| Native 10-Foot UI | **Yes (Siri Remote)** | Port only | Mobile only | Clunky |
| Price / Barrier | **Free Prototype** | $30 / pack | $10+/mo | Ad-heavy |
| 100% Offline | **Yes (Bots+JSON)** | Needs Net | Cloud only | Crashes |
| Local Heritage | **Yes (Sri Lanka)** | No | Unverified | None |

## CORE ADVANTAGES

- Zero Cost: Free prototype with no paywalls or ads
- Couch Ergonomics: First-class Siri Remote Focus Engine
- Offline Autonomy: Never fails when Wi-Fi is spotty
- Culturally Rich: First dedicated Sri Lankan trivia on tvOS

## PHASED ROADMAP

### PHASE 1: IMPLEMENTED (CURRENT)

- Native tvOS 17 app in Swift 5.9
- 6 categories with 42 verified questions
- Siri Remote focus & offline bot simulation
- 10/10 passing unit tests & JSON storage

### PHASE 2: PROPOSED COMPANION (NEXT)

- Lightweight WebSocket server
- Real-time phone buzzers via partyplay.tv/join
- Zero app install needed on phones

### PHASE 3: FUTURE ECOSYSTEM

- FaceTime SharePlay for remote game nights
- Custom trivia question creator portal
- Siri voice-command answer support

TECHNICAL IMPLEMENTATION

# Native Architecture & Verified Quality

Clean MVVM architecture, 100% native Apple frameworks, and verified automated tests.

## CORE ARCHITECTURE & TECH STACK

**▶ Swift 5.9 & tvOS 17.2 SDK**
Built in Xcode 15.2 with under 15MB binary size.

**▶ Strict MVVM Pattern**
Centralized AppContainer manages service dependency injection.

**▶ Combine State Machine**
Coordinates countdown timers, input locks, and score calculations.

**▶ CoreImage Vector QR**
Native CIFilter draws sharp QR codes on-the-fly without 3rd-party pods.

**▶ 0 External Dependencies**
100% native Apple frameworks for maximum stability and speed.

## VERIFIED UNIT TESTING & METRICS

**✓ 10/10 Passing Unit Tests (100% Pass Rate)**
- ScoreServiceTests (4/4): Base points, speed bonuses, tie-breaks
- QuizDataServiceTests (4/4): JSON integrity, category filters, Sri Lanka content
- GameEngineTests (3/3): Clean reset, single-click lock, game loop

**✓ Verified Performance**
- Fluid 60 FPS remote focus navigation on Apple TV 4K
- Zero memory leaks across complete match cycles
- Full Reduce Motion & High Contrast support

**✓ Practical Impact**
- Makes television an active social medium for the living room
- Delivers cultural knowledge in a fun, competitive format

CONCLUSION & DEMONSTRATION

# Bringing the Game Show to Every Living Room

A complete, production-ready tvOS platform ready for live simulator demonstration.

INNOVATION

## Living Room Stage

Replaces passive video and solitary phone screens with a broadcast game show.

TECHNICAL RIGOR

## Native MVVM & Tests

Swift 5.9, tvOS 17 SDK, 10/10 passing unit tests, and 0 third-party bloat.

ZERO BARRIERS

## Instant Offline Play

100% offline out-of-the-box with smart AI bots and QR companion pairing.

## PROCEEDING TO LIVE SIMULATOR DEMO ON APPLE TV 4K

Live Demonstration Path:

1. Home Stage: Hero game show branding & Siri Remote focus scaling
2. Category Select: Browse the 6 themes and select the 'Sri Lanka Quiz'
3. Party Lobby: Room code PLAY88, QR generator & simulated bots
4. Quiz Arena: Circular countdown ring, speed bonus (+50%) & educational context
5. Champion Podium: 3D-style podium celebration and accuracy rankings

**Thank you for your time. Presenter:** Jayasundara R.K.M.J.Y. (IT22136374) • SE4041 MADD Part B